Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table? 2

show ip route OSPF

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you? 2

It tells us that because interface S0/0/1 on R2 is still configured as a passive interface, no OSPF Hello packets are sent out of that interface. Therefore, R2 and R3 cannot form an OSPF neighbor adjacency, forcing R3 to learn the route to R2's network indirectly via R1. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

``text

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost is 65. It is calculated by adding the OSPF cost of the outgoing interface on R3 (Serial0/0/1, cost = 64) and the destination outgoing interface on R2 (GigabitEthernet0/0, cost = 1). 2

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default OSPF reference bandwidth is 100 Mbps. Because the OSPF cost formula is (Reference Bandwidth / Interface Bandwidth), any interface with a speed of 100 Mbps or faster will result in a cost of 1. Increasing the reference bandwidth allows the OSPF routing process to accurately calculate metrics and differentiate between these higher-speed links.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The accumulated OSPF cost is calculated by summing up the costs of all the **outgoing** interfaces along the routing path to the destination network. The cost for each individual interface is determined by the formula: 100,000,000 / bandwidth_in_bps. For example, a default serial interface (1.544 Mbps) has a cost of 64, while a GigabitEthernet interface has a cost of 1.

1

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost is ~~**781**~~. This is because the bandwidth 128 command manually sets the interface bandwidth to 128 kbps (128,000 bps). Using the OSPF cost formula (100,000,000 / 128,000), the resulting cost for that single outgoing serial interface becomes 781.

0

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The route path changed because the OSPF cost of the direct serial link between R1 and R3 was manually increased using the ip ospf cost command. As a result, the direct path's accumulated cost metric is now higher than the indirect path traversing through R2. Since the OSPF algorithm always prefers the route with the lowest accumulated cost, it dynamically recalculates and chooses the path through R2.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Controlling the router ID is important because it uniquely identifies the router within the OSPF domain. A strategically assigned router ID makes it much easier to manage the network and troubleshoot routing issues. Additionally, in multi-access networks, the router ID determines which router becomes the Designated Router (DR) or Backup Designated Router (BDR) when interface priorities are equal.

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2. Why is the DR/BDR election process not a concern in this lab?

   The DR/BDR election process is not a concern in this lab because the routers are connected to each other using point-to-point serial links. DR and BDR elections only occur on multi-access networks (like Ethernet). On a point-to-point link, there are only two devices, so a DR/BDR is unnecessary.

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3. Why would you want to set an OSPF interface to passive?

   Setting an interface to passive prevents OSPF routing updates (Hello packets) from being sent out of that interface, while still allowing the network associated with that interface to be advertised to other OSPF routers. This is typically used for LAN interfaces connected to end devices (like PCs) to save network bandwidth, reduce processing load on end devices, and improve security by preventing unauthorized devices from forming OSPF adjacencies.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of "Areas" (such as Area 0, the backbone area) is the aspect of OSPF that makes it hierarchical. By dividing a large OSPF domain into multiple smaller areas, OSPF restricts the flooding of link-state updates to within an area, reduces the size of routing tables, and limits the impact of network topology changes. This area-based structure allows OSPF networks to be highly scalable.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The route from **EIGRP** will be used. When a router receives routes to the exact same destination network from different routing protocols, it ignores the metrics and instead uses the Administrative Distance (AD) to determine the most reliable route source. EIGRP has an AD of 90, OSPF has an AD of 110, and RIP has an AD of 120. Since EIGRP has the lowest AD, its route is placed into the routing table.

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